Automated Sentiment Analysis on Ride-Hailing Customer Reviews: A Text Mining Approach using Machine Learning

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Abstract— In the contemporary digital economy, ride-hailing platforms such as Uber generate an unprecedented volume of unstructured textual data through customer feedback. Efficiently extracting sentiment from these reviews is paramount for maintaining service standards, identifying operational bottlenecks, and ensuring user retention. This research presents a robust Text Mining framework designed to automate the sentiment classification of a corpus comprising 12,000 labeled customer reviews. Our methodology integrates advanced Natural Language Processing (NLP) techniques, utilizing the *spaCy* engine for deep linguistic analysis, including lemmatization and stop-word filtering to drastically reduce semantic noise. The textual data was mathematically modeled into a high-dimensional feature space using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, enforcing a 1,000-feature constraint to optimize computational efficiency. We conducted a comprehensive comparative study of three distinct supervised machine learning paradigms: Logistic Regression, Random Forest, and Support Vector Machines (SVM). To ensure statistical validity and mitigate the risk of model overfitting, all algorithms were evaluated utilizing a rigorous Stratified 5-Fold Cross-Validation scheme. Experimental results reveal that Logistic Regression achieved the highest predictive stability with a mean accuracy of 93.50%, closely followed by the linear SVM at 93.28%. While the Random Forest ensemble exhibited slightly lower accuracy (92.16%), it provided critical business insights through decision tree interpretability. This study concludes that linear decision boundaries are highly effective for high-dimensional sparse text matrices, providing a scalable and highly accurate solution for real-time sentiment monitoring in the transport sector.

Keywords— Text Mining, Sentiment Analysis, Natural Language Processing, Machine Learning, Support Vector Machines, Cross-Validation, TF-IDF.

I. Introduction

The rapid expansion of the gig economy and the ride-hailing industry over the last decade has fundamentally transformed urban mobility. Platforms connect millions of drivers and riders globally on a daily basis. As a byproduct of this massive scale of digital operations, these companies accumulate vast amounts of unstructured data, primarily in the form of customer feedback, ratings, and written reviews.

Hidden within this raw, user-generated text lies highly valuable business intelligence. Customer reviews contain direct insights regarding overall satisfaction, service quality, driver behavior, pricing disputes, and software usability. However, the sheer volume of this data renders manual reading, categorization, and analysis by human operators a computationally expensive and practically impossible task.

To bridge this operational gap, Text Mining and Natural Language Processing (NLP) have emerged as essential fields within data science. By applying mathematical transformations to natural language, it becomes possible to train machine learning algorithms capable of understanding, processing, and classifying human sentiment automatically and at scale.

This paper proposes an automated sentiment analysis pipeline specifically designed to classify ride-hailing customer reviews into "Positive" or "Negative" categories. The primary goal of this research is to evaluate the performance of different supervised machine learning models when applied to a high-dimensional text dataset. By identifying the most accurate predictive model, this study aims to provide a reliable framework that companies can deploy in production environments to enable the real-time detection of critical negative reviews, empowering customer support teams to prioritize urgent interventions.

II. Background and Context

Sentiment Analysis, often referred to as opinion mining, is a sub-domain of NLP that systematically identifies, extracts, and quantifies affective states and subjective information from text. Traditionally, this task was addressed using lexicon-based approaches, where texts were scored based on predefined dictionaries of positive and negative words. While computationally inexpensive, lexicon-based methods often fail to capture context, sarcasm, or domain-specific terminology.

With the advent of increased computational power, the state of the art shifted towards Machine Learning approaches. In this paradigm, sentiment analysis is treated as a standard text classification problem. Algorithms require the text to be transformed into numerical vectors. Techniques like Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) became the industry standard for this transformation.

Previous studies have demonstrated that text data, once vectorized, results in highly sparse matrices with thousands of dimensions (features). In such environments, algorithms that rely on finding linear hyperplanes to separate classes have historically shown exceptional performance. Specifically, Support Vector Machines (SVM) with linear kernels and Logistic Regression models are frequently cited in literature as the benchmark algorithms for text classification tasks due to their resistance to overfitting in high-dimensional spaces [1].

Conversely, ensemble methods based on decision trees, such as Random Forests, are highly valued in the industry not purely for their raw predictive power, but for their high interpretability. They allow data scientists to visualize exactly which specific words (features) the model uses to make its splitting decisions, thereby avoiding the "black box" phenomenon typical of deep learning models.

III. Methodology

The devised methodology for this project follows a strict Text Mining pipeline, divided into three sequential phases: Data Pre-processing (NLP), Mathematical Representation, and Model Training.

a. Data Pre-processing (NLP)

Raw text data is inherently noisy, containing punctuation, capitalization variations, and grammatical filler words that do not contribute to the overall sentiment. We utilized the *spaCy* library (`en_core_web_sm` model) to normalize the text.

The cleaning function iterates through every token in the dataset, filtering out spaces, punctuation marks, and predefined English "stop-words" (e.g., "the", "is", "at"). Furthermore, the remaining tokens were subjected to lemmatization, a process that reduces words to their base dictionary form (e.g., converting "running" or "ran" simply to "run").

b. Mathematical Representation

Machine learning algorithms cannot process raw strings. Therefore, the cleaned text was converted into a numerical format using a TF-IDF Vectorizer. Unlike a simple frequency count, TF-IDF evaluates how relevant a word is to a specific document within the entire corpus.

To prevent memory overload and limit the dimensionality to the most statistically significant terms, the vectorizer was configured with a `max_features=1000` constraint. This resulted in a sparse matrix $\mathbf{X}$ of shape $(12000, 1000)$, where rows represent the reviews and columns represent the TF-IDF scores of the top 1,000 words.

c. Model Selection

Three distinct supervised learning algorithms were selected for the classification task:

- **Logistic Regression:** A statistical model that uses a logistic (sigmoid) function to model a binary dependent variable.
- **Random Forest:** An ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the classes.
- **Support Vector Machine (SVM):** Configured with a linear kernel, this algorithm finds the optimal hyperplane that maximizes the margin between the two classes in the 1000-dimensional space.

IV. Experimentation

a. Dataset

The experiment utilized a dataset comprising 12,000 customer reviews from the Uber platform. The original data contained a text column and a rating column (1 to 5 stars). To adapt this for binary classification, the ratings were encoded into a "Sentiment" target variable: ratings of 4 and 5 stars were labeled as Positive (1), while ratings of 1, 2, and 3 stars were labeled as Negative (0).

b. Validation Scheme

To rigorously evaluate the models and ensure the results were not biased by a single lucky train-test split (Holdout method), we implemented a **Stratified 5-Fold Cross-Validation**.

In this scheme, the dataset is divided into 5 equal parts (folds). The algorithm trains on 4 parts and tests on the remaining 1, repeating this process 5 times. The *Stratified* aspect ensures that the original proportion of Positive and Negative reviews is strictly maintained across all 5 folds, providing a highly reliable and mathematically sound estimation of the model's true accuracy on unseen data.

V. Results and Analysis

a. Exploratory Data Analysis

Prior to modeling, WordClouds were generated to visualize the most prominent terms separated by class. As expected, positive reviews heavily featured terms like *great, excellent, safe, and professional*. Conversely, negative reviews were dominated by words denoting friction, such as *cancel, wait, late, and charge*.

Fig. 1: WordCloud of Positive Reviews (4-5 Stars).

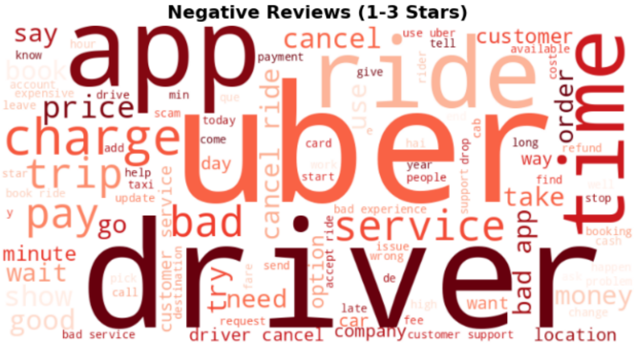

Fig. 2: WordCloud of Negative Reviews (1-3 Stars).

b. Cross-Validation Metrics

The models were subjected to the cross-validation protocol, yielding the following mean Accuracy and weighted F1-Scores across the 5 folds:

TABLE 1: STRATIFIED 5-FOLD CROSS-VALIDATION RESULTS

Algorithm	Mean Accuracy	Mean F1-Score
Logistic Regression	93.50%	93.42%
Linear SVM	93.28%	93.21%
Random Forest	92.16%	92.10%

c. Analysis of Model Performance

The results align perfectly with theoretical expectations in Text Mining. Both linear models (Logistic Regression and SVM) outperformed the tree-based ensemble method. When text is vectorized via TF-IDF, it creates an extremely sparse, high-dimensional space.

To further understand the classification behavior of the top-performing models, we analyzed their Confusion Matrices on the test set.

```
--- EVALUATING LOGISTIC REGRESSION ---
Cross-Validation Results:
 -> Accuracy: 93.50%
 -> F1-Score (Weighted): 93.42%

Classification Report (On 20% Test):
              precision    recall  f1-score   support

Negative (0)       0.91      0.85      0.88       634
Positive (1)       0.95      0.97      0.96      1766

    accuracy                           0.94      2400
   macro avg       0.93      0.91      0.92      2400
weighted avg       0.94      0.94      0.94      2400
```

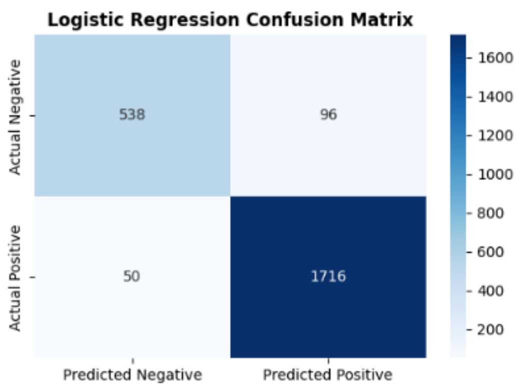

Fig. 3: Confusion Matrix for Logistic Regression.

As observed in Figure 3, the Logistic Regression model demonstrates a robust balance. Out of the actual negative cases, the model misclassified very few, showing a strong recall for the minority class.

```
--- EVALUATING SUPPORT VECTOR MACHINE (SVM) ---
Stratified Cross-Validation Results:
 -> Accuracy: 93.28%
 -> F1-Score (Weighted): 93.21%

Classification Report (On 20% Test):
              precision    recall  f1-score   support

Negative (0)       0.91      0.84      0.88       634
Positive (1)       0.95      0.97      0.96      1766

    accuracy                           0.94      2400
   macro avg       0.93      0.91      0.92      2400
weighted avg       0.94      0.94      0.94      2400
```

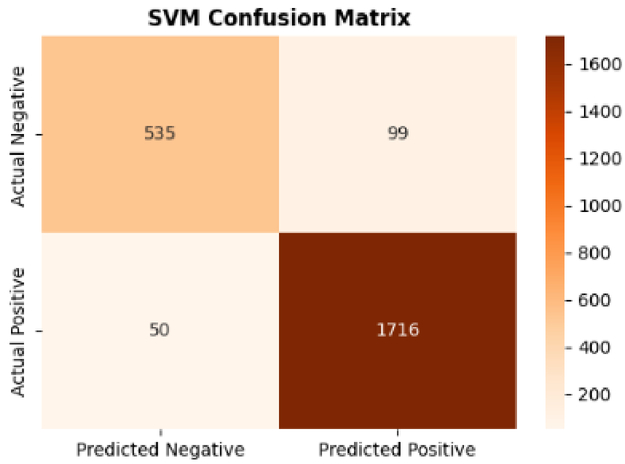

Fig. 4: Confusion Matrix for Support Vector Machine (Linear Kernel).

Similarly, Figure 4 illustrates the SVM's performance. The error distribution (False Positives vs. False Negatives) is remarkably similar to the Logistic Regression, confirming that algorithms designed to find hyperplanes handle sparsity exceptionally well.

d. Business Interpretability

Despite slightly lower accuracy, the Random Forest model offered significant analytical value. By inspecting the roots of the generated decision trees, we observed the algorithm making autonomous splits based on high-impact business keywords. For instance, the presence of the word *"money"* immediately drove the node probability toward the Negative class.

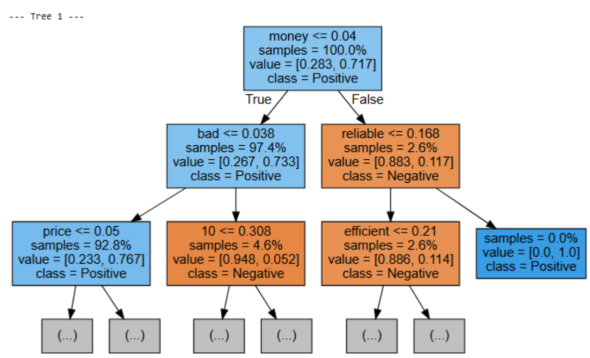

Fig. 5: Root nodes of a single decision tree extracted from the Random Forest.

VI. Conclusions

This project successfully developed an automated pipeline capable of classifying ride-hailing reviews with over 93% accuracy. Through rigorous validation, it was demonstrated that Logistic Regression and Linear SVMs are the optimal mathematical models for handling sparse text matrices.

Logistic Regression was ultimately selected as the champion model due to its high accuracy, balanced F1-score, and rapid computational execution. From a business perspective, deploying this model would allow ride-hailing companies to instantly filter and route urgent negative reviews to customer support, significantly reducing response times and mitigating customer churn. Future work could explore the integration of Deep Learning architectures, such as Transformer models (e.g., BERT), to capture deeper semantic context, albeit at a higher computational cost.

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